GCSE Maths

Complete Revision Summary Sheets

Foundation & Higher Tier · All Topics

1. Number
2. Algebra
3. Ratio & Proportion
4. Geometry
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6. Graphs
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9. Further Algebra ★

★ Higher Tier content clearly labelled throughout

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1 · Number

Integers, fractions, decimals, percentages, powers, roots & standard form

F Foundation | H Higher

TYPES OF NUMBER

- **Natural numbers:** 1, 2, 3, 4, ...
- **Integers:** ...−2, −1, 0, 1, 2, ...
- **Rational:** expressible as p/q (p,q integers)
- **Irrational:** cannot be written as fraction (e.g. √2, π)
- **Prime:** exactly 2 factors: 1 and itself
- **Square numbers:** 1, 4, 9, 16, 25, 36, 49, 64, 81, 100
- **Cube numbers:** 1, 8, 27, 64, 125

FACTORS, MULTIPLES & PRIMES

- **HCF** (Highest Common Factor): largest number that divides both
- **LCM** (Lowest Common Multiple): smallest number both divide into
- **Prime factorisation:** write as product of primes

```
360 = 2³ × 3² × 5
HCF(36,48): 2² × 3 = 12
LCM(4,6): 12
```

% FRACTIONS, DECIMALS & PERCENTAGES

```
Fraction → Decimal: divide numerator ÷ denominator
Decimal → %: multiply by 100
% of amount: (% ÷ 100) × amount
% change: (change ÷ original) × 100
Reverse %: amount ÷ (1 ± %/100)
```

EXAMPLE – REVERSE %

After 20% increase, price = £144. Find original.

144 ÷ 1.20 = **£120**

⚡ POWERS, ROOTS & STANDARD FORM

```
aᵐ × aⁿ = aᵐ⁺ⁿ   (multiply → add powers)
aᵐ ÷ aⁿ = aᵐ⁻ⁿ   (divide → subtract)
(aᵐ)ⁿ = aᵐⁿ       (power of power → multiply)
a⁰ = 1             a⁻ⁿ = 1/aⁿ
a^(1/n) = ⁿ√a      a^(m/n) = (ⁿ√a)ᵐ
```

Standard form: $A \times 10^n$ where $1 \le A < 10$

```
3500 = 3.5 × 10³
```

★ Higher Tier — Surds

√ SURD RULES

```
√a × √b = √(ab)
√a ÷ √b = √(a/b)
(√a)² = a
(a + √b)(a - √b) = a² - b
```

Simplify: find largest perfect-square factor

```
√72 = √(36×2) = 6√2
```

÷ RATIONALISING THE DENOMINATOR

Multiply top and bottom to remove surd from denominator.

```
1/√3 × √3/√3 = √3/3
5/(2+√3) × (2-√3)/(2-√3) = (10-5√3)/(4-3) = 10-5√3
```

WORKED EXAMPLE

Simplify (3+√5)(3-√5) = 9 - 5

= 4

CALCULATOR & ESTIMATION TIPS

- Round to 1 or 2 significant figures before estimating
- Significant figures (s.f.): count from first non-zero digit
- Decimal places (d.p.): count from decimal point
- Error intervals: if x = 3.4 (1 d.p.) → 3.35 ≤ x < 3.45

2 · Algebra

Expressions, equations, inequalities, factorising & sequences

F Foundation | H Higher

✏ EXPANDING BRACKETS

```
a(b + c) = ab + ac
(x+a)(x+b) = x² + (a+b)x + ab
(a+b)² = a² + 2ab + b²
(a-b)² = a² - 2ab + b²
(a+b)(a-b) = a² - b²
```

EXAMPLE

Expand (2x+3)(x-4)

$= 2x^2 - 8x + 3x - 12$

$= 2x^2 - 5x - 12$

FACTORISING

- **Common factor:** $6x^2 + 9x = 3x(2x + 3)$
- **Difference of squares:** $x^2 - 16 = (x+4)(x-4)$
- **Trinomial:** $x^2 + 5x + 6 = (x+2)(x+3)$
- **Grouping:** $ax + ay + bx + by = (a+b)(x+y)$

EXAMPLE – FACTORISE $2X^2 + 7X + 3$

Find two numbers: 2×3=6, sum=7 → 1 and 6

$2x^2 + x + 6x + 3 = x(2x+1) + 3(2x+1)$

$= (x+3)(2x+1)$

⚖ SOLVING EQUATIONS

- Do the same operation to both sides
- Collect like terms, then isolate the variable
- **Simultaneous:** elimination or substitution

```
3x + 5 = 20  →  x = 5
Sim. equations:
  2x + y = 7
  x + y  = 4  →  x=3, y=1
```

≤ INEQUALITIES

- Solve like equations BUT flip sign when multiplying/dividing by negative
- Show on number line: ○ = open (not included), ● = closed (included)

```
3x - 2 > 7  →  3x > 9  →  x > 3
-2x ≤ 8  →  x ≥ -4  (sign flips!)
```

Integer solutions of $-1 \le x < 4$: -1, 0, 1, 2, 3

SEQUENCES

- **Arithmetic (linear):** constant difference d
- **Geometric:** constant ratio r
- **nth term (arithmetic):** a + (n-1)d
- **Quadratic nth term:** contains n^2 (Higher)

```
Sequence: 5, 8, 11, 14, …  d = 3
nth term = 3n + 2
(Check: n=1 → 5 ✓)
```

FORMULAE & REARRANGING

- Substitute known values and solve for unknown
- Change subject: inverse operations step by step

```
v = u + at  →  make t subject:
v - u = at
t = (v - u) / a
```

EXAMPLE – MAKE R SUBJECT

$A = \pi r^2 \rightarrow r^2 = A/\pi$

$r = \sqrt{(A/\pi)}$

Key Identities to Memorise: $a^2 - b^2 = (a+b)(a-b)$ | $(a \pm b)^2 = a^2 \pm 2ab + b^2$ | Sum of arithmetic sequence = n/2 × (first + last)

3 · Ratio & Proportion

Ratio, proportion, rates of change, speed-distance-time & percentage change

F Foundation | H Higher

⚖ RATIO

- Simplify by dividing by HCF: 12:8 → 3:2
- **Sharing:** find total parts, then calculate each
- **Equivalent ratios:** multiply/divide all parts by same number
- **Map scales:** 1:50 000 means 1 cm = 500 m

SHARE £240 IN RATIO 3:5

Total parts = 8 | 1 part = £30

£90 : £150

PROPORTION

- **Unitary method:** find value of 1 unit first
- **Best value:** divide price by quantity to get price per unit
- **Exchange rates:** multiply or divide consistently

```
5 pens cost £3.50
1 pen = £0.70
8 pens = £5.60
```

SPEED, DISTANCE, TIME

```
Speed = Distance ÷ Time
Distance = Speed × Time
Time = Distance ÷ Speed
```

Units: km/h, m/s, mph. Convert carefully!

DENSITY & PRESSURE

```
Density = Mass ÷ Volume
Pressure = Force ÷ Area
```

Same triangle method as SDT applies.

% CHANGE & GROWTH

```
Multiplier for n% increase: 1 + n/100
Multiplier for n% decrease: 1 - n/100
Compound interest:
A = P(1 + r/100)ⁿ
```

★ Higher Tier — Direct & Inverse Proportion

∝ DIRECT PROPORTION

- $y \propto x$ means $y = kx$
- $y \propto x^2$ means $y = kx^2$
- $y \propto \sqrt{x}$ means $y = k\sqrt{x}$
- **Find k:** substitute known values

EXAMPLE

$y \propto x^2$, y=12 when x=2 → k=3

When x=5: y = 3×25 = **75**

÷ INVERSE PROPORTION

- $y \propto 1/x$ means $y = k/x$
- $y \propto 1/x^2$ means $y = k/x^2$
- As x increases, y decreases (and vice versa)

EXAMPLE

$y \propto 1/x$, y=6 when x=4 → k=24

When x=8: y = 24/8 = **3**

4 · Geometry & Measures

Angles, shapes, area, volume, circles, transformations & vectors

F Foundation | H Higher

ANGLE RULES

- Angles on a straight line = 180°
- Angles around a point = 360°
- Vertically opposite angles are equal
- Alternate angles (Z-angles) are equal
- Co-interior angles sum to 180°
- Corresponding angles (F-angles) are equal

POLYGONS

```
Interior angle sum = (n-2) × 180°
Each interior angle (regular) = (n-2)×180 ÷ n
Each exterior angle (regular) = 360 ÷ n
Interior + Exterior = 180°
```

Triangle = 180°, Quad = 360°, Pentagon = 540°, Hexagon = 720°

AREA & PERIMETER

Shape	Area	Perimeter
Rectangle	l × w	2(l + w)
Triangle	½ × b × h	sum of sides
Parallelogram	b × h	2(a + b)
Trapezium	½(a+b) × h	sum of sides
Circle	πr^2	$2\pi r$

VOLUME & SURFACE AREA

Shape	Volume
Cuboid	l × w × h
Prism	Area of cross-section × length
Cylinder	$\pi r^2 h$
Cone	$\frac{1}{3}\pi r^2 h$
Sphere	$\frac{4}{3}\pi r^3$
Pyramid	⅓ × base area × h

TRANSFORMATIONS

- **Translation:** describe with vector (x/y)
- **Reflection:** state mirror line equation
- **Rotation:** state centre, angle, direction
- **Enlargement:** state scale factor & centre

Enlargement with SF < 1 = reduction. Negative SF = rotation through centre.

CONGRUENCE & SIMILARITY

- **Congruent:** same shape and size (SSS, SAS, ASA, RHS)
- **Similar:** same shape, different size
- Length ratio = k → Area ratio = k^2 → Volume ratio = k^3

```
Similar triangles: if sides in ratio 2:3
Areas in ratio 4:9, Volumes in ratio 8:27
```

★ Higher Tier — Circle Theorems & Vectors

CIRCLE THEOREMS

- Angle at centre = 2 × angle at circumference
- Angles in same segment are equal
- Angle in semicircle = 90°
- Opposite angles in cyclic quadrilateral = 180°
- Tangent meets radius at 90°
- Two tangents from external point are equal
- Alternate segment theorem

→ VECTORS

```
Adding:  a + b  (tip-to-tail)
Subtracting: a - b = a + (-b)
Magnitude: |a| = √(x² + y²)
Scalar multiple: k(x/y) = (kx/ky)
```

Parallel vectors: b = ka for some scalar k. Midpoint of AB = A + ½AB

5 · Trigonometry & Pythagoras

Pythagoras' theorem, SOHCAHTOA, sine & cosine rules

F Foundation | H Higher

△ PYTHAGORAS' THEOREM

Only for **right-angled** triangles.

```
a² + b² = c²   (c = hypotenuse)
c = √(a² + b²)
a = √(c² - b²)
```

FIND HYPOTENUSE: LEGS 3, 4

c = √(9 + 16) = √25

c = 5

SOH CAH TOA

Labels: **O**pposite, **A**djacent, **H**ypotenuse (relative to angle θ)

```
Sin θ = O/H   →  O = H·sinθ,  H = O/sinθ
Cos θ = A/H   →  A = H·cosθ,  H = A/cosθ
Tan θ = O/A   →  O = A·tanθ,  A = O/tanθ
```

To find angle: use inverse $\sin^{-1}$, $\cos^{-1}$, or $\tan^{-1}$

EXACT TRIGONOMETRIC VALUES (MUST KNOW)

θ	sin θ	cos θ	tan θ
0°	0	1	0
30°	1/2	√3/2	1/√3 = √3/3
45°	1/√2 = √2/2	1/√2 = √2/2	1
60°	√3/2	1/2	√3
90°	1	0	undefined

★ Higher Tier — Sine Rule, Cosine Rule & Area Formula

SINE RULE

Use when you know: **two sides & non-included angle**, or two angles & one side.

```
a/sin A = b/sin B = c/sin C
(Flip for finding angles:)
sin A/a = sin B/b = sin C/c
```

FIND SIDE B: A=40°, B=75°, A=8

b/sin75° = 8/sin40°

b = 8×sin75°/sin40° **≈ 12.0**

COSINE RULE & AREA

Use when you know: **two sides & included angle**, or all three sides.

```
a² = b² + c² - 2bc·cos A
cos A = (b² + c² - a²) / 2bc
Area = ½ab·sin C
```

AREA: A=6, B=9, C=35°

Area = ½ × 6 × 9 × sin35°

≈ 15.5 cm²

6 · Graphs & Functions

Straight lines, curves, sketching, functions & graph transformations

F Foundation | H Higher

STRAIGHT LINE GRAPHS

```
y = mx + c
  m = gradient = rise/run = (y₂-y₁)/(x₂-x₁)
  c = y-intercept
Parallel lines: same gradient
Perpendicular: gradients multiply to -1
```

LINE THROUGH (1,3) AND (3,7)

m = (7-3)/(3-1) = 4/2 = 2

y - 3 = 2(x - 1) → **y = 2x + 1**

READING GRAPHS

- **Distance-time:** gradient = speed
- **Velocity-time:** gradient = acceleration; area = distance
- **Real-life graphs:** identify intercepts and gradients in context
- **Conversion graphs:** read off and interpolate

★ Higher Tier — Quadratic, Cubic & Other Graphs

U QUADRATIC GRAPHS: $Y = AX^2 + BX + C$

- a > 0 → U-shape; a < 0 → ∩-shape
- Roots: where graph crosses x-axis (y=0)
- y-intercept: at x=0, y=c
- Line of symmetry: x = -b/(2a)
- Vertex (turning point): substitute x back in

∿ OTHER GRAPH SHAPES

Equation	Shape
$y = x^3$	Cubic (S-curve)
y = 1/x	Reciprocal (two curves)
$y = a^x$	Exponential (a>1 = growth)
y = sin x	Wave, period 360°
y = cos x	Wave, period 360°

★ Higher Tier — Functions & Graph Transformations

ƒ FUNCTIONS

```
f(x): substitute x into expression
fg(x): apply g first, then f
f⁻¹(x): inverse function (swap x and y, rearrange)
Domain: set of allowed inputs
Range: set of possible outputs
```

↔ GRAPH TRANSFORMATIONS

Transformation	Effect
f(x) + a	Translate up by a
f(x + a)	Translate left by a
af(x)	Stretch vertically × a
f(ax)	Stretch horizontally × 1/a
-f(x)	Reflect in x-axis
f(-x)	Reflect in y-axis

7 · Probability

Basic probability, combined events, tree diagrams & Venn diagrams

F Foundation | H Higher

BASIC PROBABILITY

```
P(event) = number of favourable outcomes / total outcomes
P(A') = 1 - P(A)   (complement)
0 ≤ P(A) ≤ 1 always
P(certain event) = 1
P(impossible event) = 0
```

Relative frequency ≈ probability (with large number of trials)

COMBINED EVENTS

```
P(A or B) = P(A) + P(B) - P(A and B)
Mutually exclusive: P(A or B) = P(A) + P(B)
Independent:   P(A and B) = P(A) × P(B)
```

P(RED) = 0.3, P(BLUE) = 0.5 (MUTUALLY EXCLUSIVE)

P(Red or Blue) = 0.3 + 0.5 = **0.8**

TREE DIAGRAMS

- Multiply along branches (AND)
- Add end probabilities for same outcome (OR)
- All branches from a point must sum to 1
- Use for 2 or more events in sequence

P(HH) FROM TWO FAIR COIN TOSSES

P(H) × P(H) = ½ × ½ = **¼**

VENN DIAGRAMS

- A ∪ B = A or B (union)
- A ∩ B = A and B (intersection)
- A' = not A (complement)
- ξ = universal set (rectangle)

```
P(A∪B) = P(A) + P(B) - P(A∩B)
```

★ Higher Tier — Conditional Probability

CONDITIONAL PROBABILITY

```
P(A | B) = P(A ∩ B) / P(B)
"Probability of A given B has occurred"
Dependent events: P(A and B) = P(A) × P(B|A)
```

EXAMPLE

Bag: 4 red, 6 blue. Draw 2 without replacement.

P(RR) = 4/10 × 3/9 = 12/90 = **2/15**

FREQUENCY TREES

- Like tree diagrams but with actual frequencies
- Each branch shows a count, not a probability
- Use to estimate probabilities from data

```
Estimated P(A) = frequency of A / total frequency
```

8 · Statistics

Averages, spread, charts, scatter graphs & cumulative frequency

F Foundation | H Higher

AVERAGES & SPREAD

Measure	How to find
Mean	Sum ÷ number of values
Median	Middle value (ordered list)
Mode	Most frequent value
Range	Max − Min
IQR	Q3 − Q1

For even-number dataset: median = average of two middle values

MEAN FROM FREQUENCY TABLE

```
Mean = Σ(fx) / Σf
  where f = frequency, x = value
Grouped data: use midpoints for x
Modal class: class with highest frequency
Median class: where cumulative freq
  reaches n/2
```

X: 2, 3, 4 | F: 3, 5, 2

Σfx = 6+15+8=29 | Σf=10

Mean = 29/10 = **2.9**

TYPES OF CHART

Chart	Used for
Bar chart	Discrete or categorical data
Pie chart	Proportions (angles sum to 360°)
Histogram	Continuous grouped data (area = frequency)
Frequency polygon	Comparing distributions
Stem & leaf	Show all data values, retains detail

SCATTER GRAPHS & CORRELATION

- **Positive correlation:** both variables increase
- **Negative correlation:** one increases, other decreases
- **No correlation:** no obvious pattern
- **Line of best fit:** drawn through the "middle" of points
- **Interpolation vs Extrapolation:** interpolation is more reliable
- Correlation ≠ causation!

★ Higher Tier — Cumulative Frequency & Box Plots

CUMULATIVE FREQUENCY

- Running total of frequencies
- Plot at **upper class boundaries**
- S-shaped (ogive) curve
- Median = value at n/2
- Q1 = value at n/4; Q3 = value at 3n/4
- IQR = Q3 − Q1

BOX PLOTS (BOX-AND-WHISKER)

- Shows: Min, Q1, Median, Q3, Max
- Box = IQR (middle 50% of data)
- Whiskers extend to Min and Max
- Compare median position and box width for distribution shape
- Skewed data: median closer to one end

⚠ **SAMPLING & DATA COLLECTION**

- **Random sampling:** every member has equal chance of selection — reduces bias
- **Stratified sampling:** divide into groups, sample in proportion to group size
- **Systematic:** every nth person
- **Biased sample:** not representative of the population

9 · Further Algebra

Quadratics in depth, completing the square, algebraic fractions, proof & iteration

H Higher Tier Only

THE QUADRATIC FORMULA

Solves $ax^2 + bx + c = 0$ for any quadratic.

```
x = (-b ± √(b² - 4ac)) / 2a
Discriminant: Δ = b² - 4ac
  Δ > 0 → 2 real roots
  Δ = 0 → 1 repeated root
  Δ < 0 → no real roots
```

SOLVE $2X^2 - 5X - 3 = 0$

a=2, b=-5, c=-3 | Δ = 25+24 = 49

x = (5 ± 7)/4 → **x = 3 or x = -½**

COMPLETING THE SQUARE

```
ax² + bx + c → a(x + b/2a)² + (c - b²/4a)
For x² + bx + c:
  (x + b/2)² - (b/2)² + c
```

COMPLETE THE SQUARE: $X^2 + 6X + 2$

$(x + 3)^2 - 9 + 2$

$(x + 3)^2 - 7$

Min value: -7 at x = -3

÷ ALGEBRAIC FRACTIONS

```
Simplify: cancel common factors (factorise first!)
Add/subtract: find common denominator
Multiply: multiply tops, multiply bottoms
Divide: multiply by reciprocal
```

SIMPLIFY $(X^2-4)/(X^2+4X+4)$

$= (x+2)(x-2) / (x+2)^2$

$= (x-2)/(x+2)$

ITERATION

- Used to find approximate solutions to equations
- Rearrange to form $x_{n+1} = f(x_n)$
- Start with initial estimate x_0
- Repeat until values converge to required accuracy

```
e.g. x³ - 2x - 3 = 0
xₙ₊₁ = (2xₙ + 3)^(1/3)
x₀=1 → x₁=1.71 → x₂=1.83 → …
```

PROOF

- **Odd integer:** $2n + 1$
- **Even integer:** $2n$
- **Consecutive integers:** $n, n+1, n+2$
- **Consecutive even:** $2n, 2n+2, 2n+4$
- **Disproof:** find a counter-example

PROVE: ODD + ODD = EVEN

$(2m+1) + (2n+1) = 2m + 2n + 2 = 2(m+n+1)$

This is even. ■

SIMULTANEOUS EQUATIONS (QUADRATIC)

One linear, one quadratic — use substitution.

```
y = x + 1  and  y = x²
Sub: x² = x + 1
x² - x - 1 = 0 → solve with formula
```

$Y = X+3$, $Y = X^2+X+1$

$x+3 = x^2+x+1 \rightarrow x^2-2 = 0$

$x = \pm\sqrt{2}$

Higher Tier Quick Summary: Quadratic formula — memorise it! | Completing the square gives vertex of parabola | Algebraic fractions — always factorise before cancelling | Proof — use algebra with n to represent general integers